

Project Proposal

**Quantum App for Nuclear Physics Integrated
into the QCOL Open Course Ecosystem**



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Project Title

Development of the First Quantum App for Nuclear Physics, Integrated into the QCOL Open Course Open Quantum Ecosystem

Project Summary

This project proposes the creation of the first specific tool designed to apply quantum computing to Nuclear Physics problems through an accessible Quantum App, integrated into the QCOL Open Course ecosystem. The application will allow simulating the time evolution of simple nuclear Hamiltonians using variational quantum algorithms, without requiring the user to program or have knowledge of quantum computing.

The Quantum App will be available within QCOL Open Course, an open, neutral, and mass-use oriented environment, where any researcher or student will be able to execute quantum simulations without technical restrictions. This project demonstrates the potential of a new generation of simple quantum applications, designed to drive the mass adoption of quantum computing.

Motivation

The use of quantum computing in Nuclear Physics exists mainly in the form of academic articles and highly specialized code. There are no tools that allow a nuclear physicist to use quantum algorithms without programming.

This project fills that gap by creating:

- the first Quantum App specifically oriented to Nuclear Physics.
- integrated into an open ecosystem (QCOL Open Course).
- designed for mass use.
- accessible to any nuclear science expert without quantum knowledge.

General Objective

Develop an accessible Quantum App, integrated into QCOL Open Course, that allows simulating time-dependent Hamiltonians of nuclear systems using variational quantum algorithms, without requiring programming knowledge or quantum computing expertise.

Specific Objectives

- Formulate a time-dependent nuclear Hamiltonian suitable for quantum simulation.
- Implement a variational quantum algorithm (VQE/VTE) to model its evolution.
- Validate results against analytical or classical numerical solutions.
- Design an accessible interface for nuclear physicists without quantum knowledge.
- Integrate the app into QCOL Open Course as an open tool for mass use.
- Demonstrate the potential of a new generation of simple Quantum Apps oriented toward mass adoption.

Innovative Approach

This project introduces three innovations that currently do not exist in the quantum ecosystem:

- First specific tool for nuclear physics based on quantum computing.
- Accessible Quantum App, with no programming required.
- Integration into an open and massive ecosystem (QCOL Open Course), available without restrictions.

Expected Results

- Functional quantum model of the selected nuclear system.
- Variational circuit capable of simulating time evolution.
- Quantum App with intuitive interface.
- Integration within the QCOL Open Course ecosystem.
- Clear and accessible documentation.

Expected Impact

- Direct access to quantum simulations without technical barriers.
- Demonstration of domain-specific quantum apps.
- Validation of QCOL as an open infrastructure for mass adoption.
- Example of a new generation of simple and accessible quantum apps.

Execution Plan (6 weeks)

Phase 1 – Theoretical Problem Statement (2 weeks)

Objective: Define the nuclear system and its time-dependent Hamiltonian.

Tasks:

- Selection of the nuclear system.
- Formulation of the time-dependent Hamiltonian.
- Mapping of the Hamiltonian to qubits.
- Design of the variational ansatz.
- Definition of physical parameters for the app.

Deliverables: Theoretical document, implementable Hamiltonian, ansatz design.

Phase 2 – Simulation with Variational Quantum Algorithms (2 weeks)

Objective: Implement the simulation of the time-dependent Hamiltonian.

Tasks:

- Implementation of the variational algorithm (VQE/VTE).
- Construction of the quantum circuit.
- Parameter optimization.

- Validation against analytical solutions.
- Energy, fidelity, and time evolution plots.

Deliverables: Functional notebook, circuit code, validated results.

Phase 3 – Quantum App + AI for Intelligent Analysis (2 weeks)

Objective: Create an accessible interface and also an AI module for automatic analysis.

Components:

1. Accessible Interface (Quantum App)

- Minimal interface design.
- Implementation in notebook or simple web format.
- Connection with the quantum backend.
- Workflow automation.

2. AI Integration

- Automatic analysis of results.
- Detection of relevant physical patterns.
- Explanations in natural language.
- Parameter recommendations.

3. Documentation and Testing

- Testing with simple cases.
- User manual.
- Demo video.

What is the QCOL Open Ecosystem and How Does This Project Integrate?

What is QCOL?

QCOL (Quantum Computing Open Lab) is an **open course** quantum infrastructure created so that anyone can use quantum computing without programming. Its mission is to democratize quantum access through:

- **Quantum Apps:** quantum applications with a simple classical interface.
- **Open Ecosystem (QCOL Open Course):** available for mass use without restrictions.
- **Universal Integration:** connects hardware, software, algorithms, AI, and communities under a unified standard.

Project Integration

The project will be integrated into **QCOL Open Course**, the open version of the QCOL ecosystem, accessible to any user without the need for programming.

Project Leader

Franci Laffita Camargo

Founder of QCOL · Nuclear Physics Researcher · Deep Tech Entrepreneur

Role within the Team

Franci is the conceptual architect and project director, responsible for:

- the physical design of the nuclear model.
- the quantum architecture of the algorithm.
- the integration with QCOL.
- the vision for mass adoption.

Professional Summary

Franci Laffita Camargo is a Nuclear Physics student at InSTEC and founder of QCOL, the first open core quantum infrastructure oriented toward mass adoption. He has developed pioneering standards such as Quantum Apps, integrating hardware, software, AI, and communities into an open ecosystem. He has an international track record in physics competitions and experience in AI-assisted development, quantum architecture, and technical leadership.